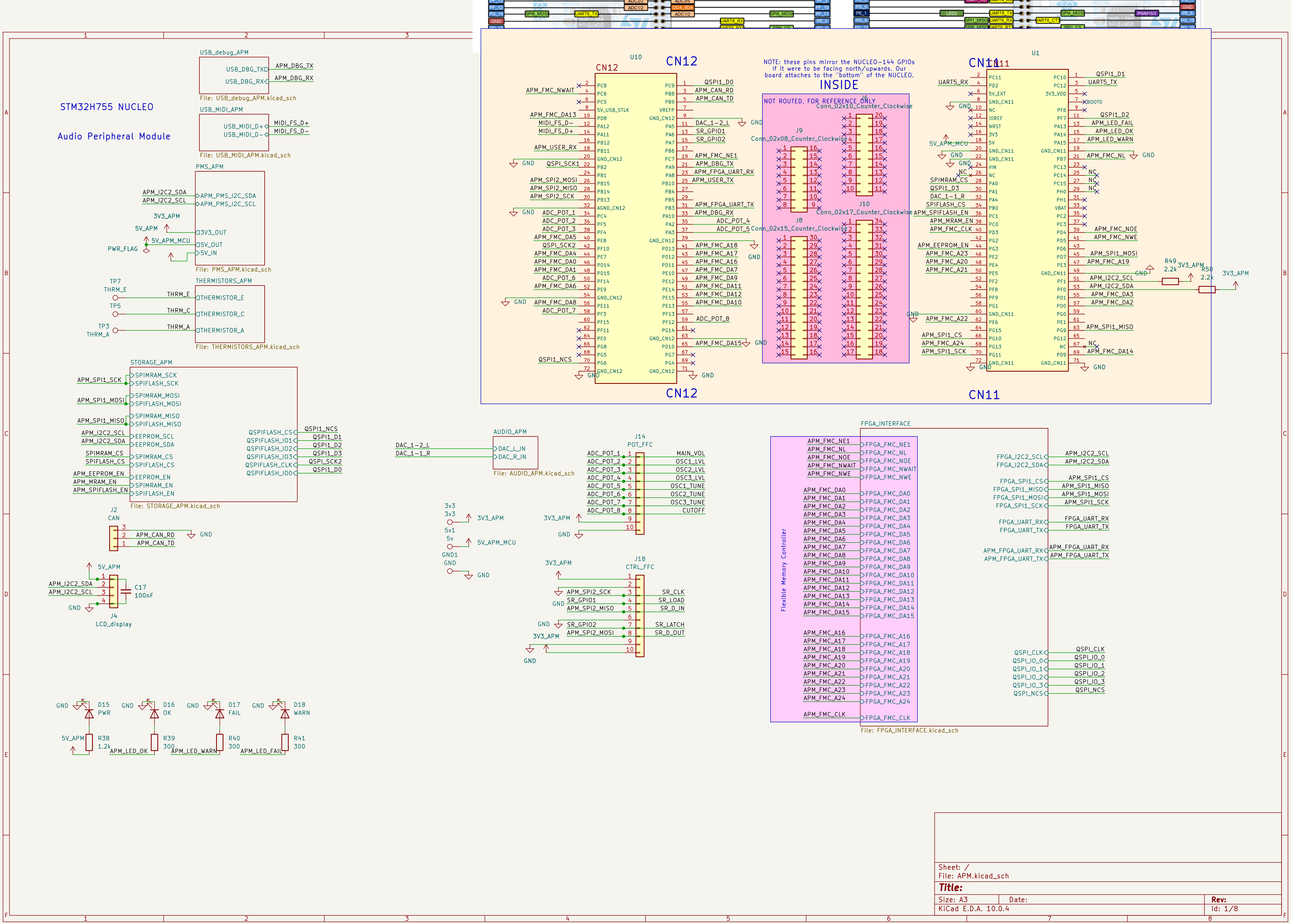
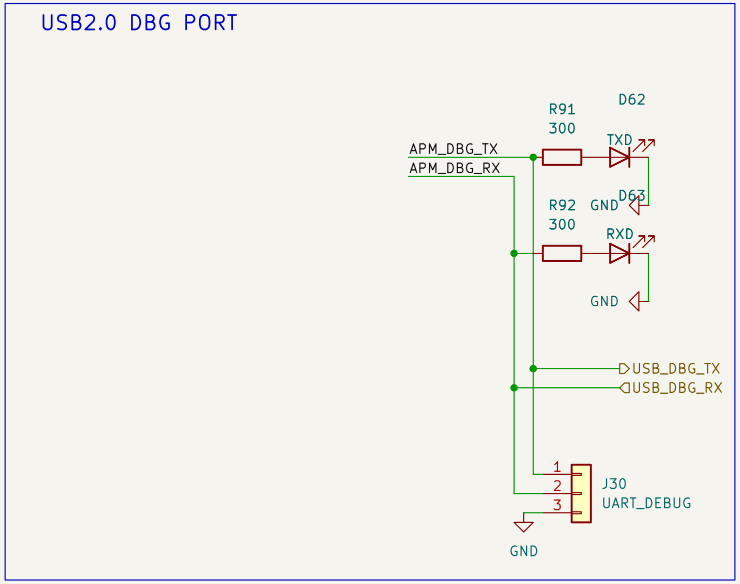

STM32H755 NUCLEO
Audio Peripheral Module
USB_debug_APM
File: USB_debug_APM.kicad_sch
USB_MIDI_APM
File: USB_MIDI_APM.kicad_sch
PMS_APM
File: PMS_APM.kicad_sch
THERMISTORS_APM
File: THERMISTORS_APM.kicad_sch
STORAGE_APM
File: STORAGE_APM.kicad_sch
AUDIO_APM
File: AUDIO_APM.kicad_sch
FPGA_INTERFACE
Flexible Memory Controller
File: FPGA_INTERFACE.kicad_sch
CN12
CN11
INSIDE
NOTE: these pins mirror the NUCLEO-144 GPIOs if it were to be facing north/upwards. Our board attaches to the "bottom" of the NUCLEO.
NOT ROUTED, FOR REFERENCE ONLY
J9 Conn_02x08_Counter_Clockwise
J6 Conn_02x10_Counter_Clockwise
J8 Conn_02x15_Counter_Clockwise
J10 Conn_02x17_Counter_Clockwise
J14 POT_FFC
J18 CTRL_FFC
J2 CAN
J4 LCD_display
C17 100nF
R49 2.2k
R50 2.2k
D15 PWR
D16 OK
D17 FAIL
D18 WARN
R38 1.2k
R39 300
R40 300
R41 300
Sheet: /
File: APM.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/8

UART -> USB Debug Port

UART debug messages avaiable via a
USB port and a 3 pin jumper for
TX, RX, and GND UART connections



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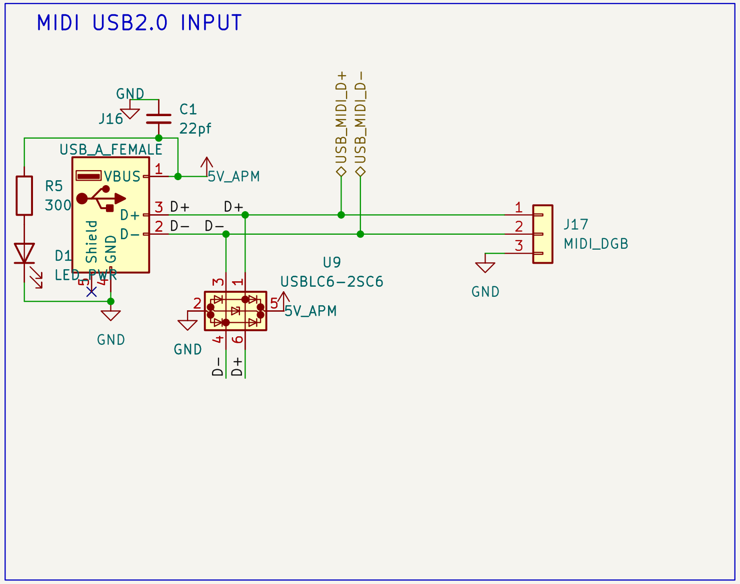
KiCad E.D.A. 10.0.4		Id: 2/8



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File: USB_MIDI_APM.kicad_sch

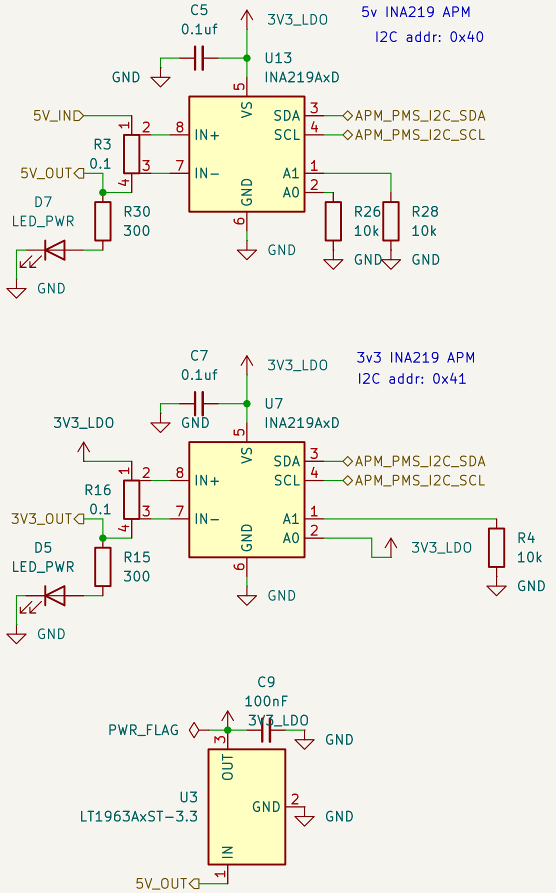
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 3/8

Power Monitor and Supply

PMS has an INA219 current sensor
for each board/node in the
PCB. Each takes in the 5V from it's
parent board and supplies 5V to said
board's circuit (its peripherals)
thus enabling monitoring



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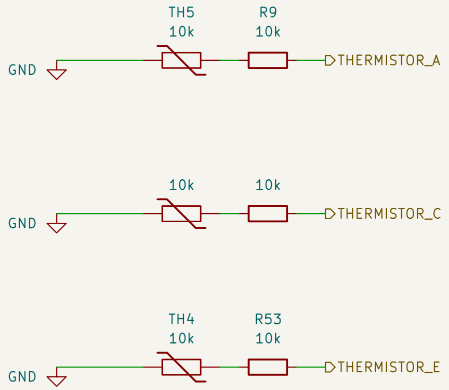
KiCad E.D.A. 10.0.4 | Id: 4/8



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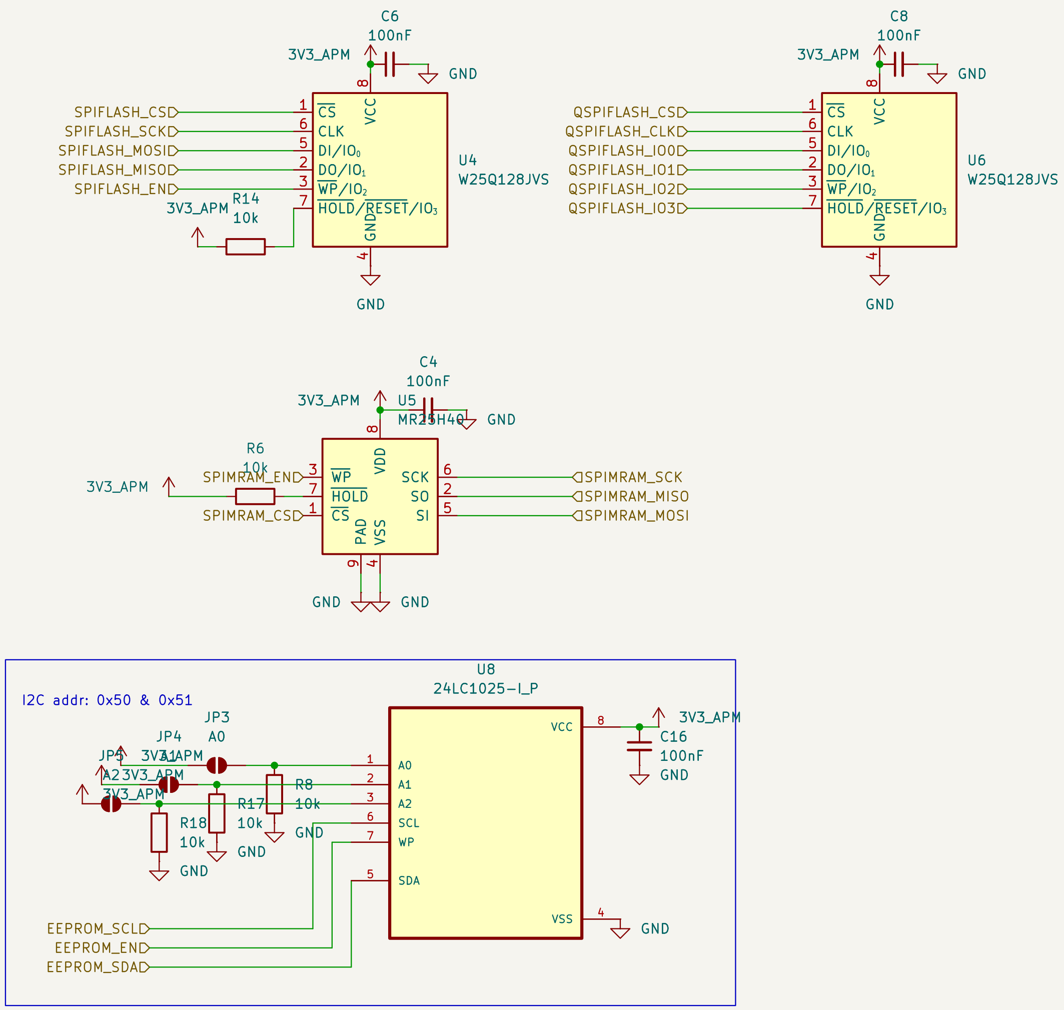
File: THERMISTORS_APM.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 5/8



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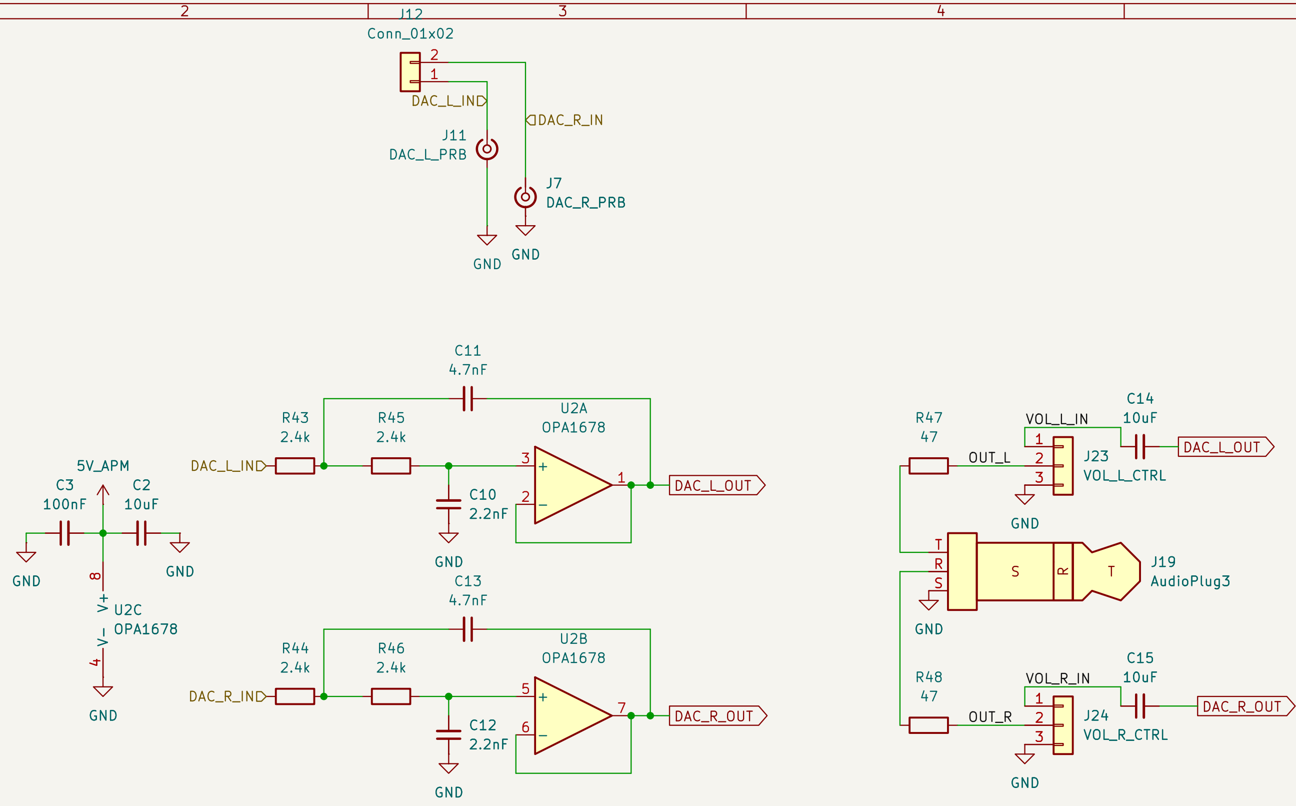
Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 6/8



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File: AUDIO_APM.kicad_sch

Title:

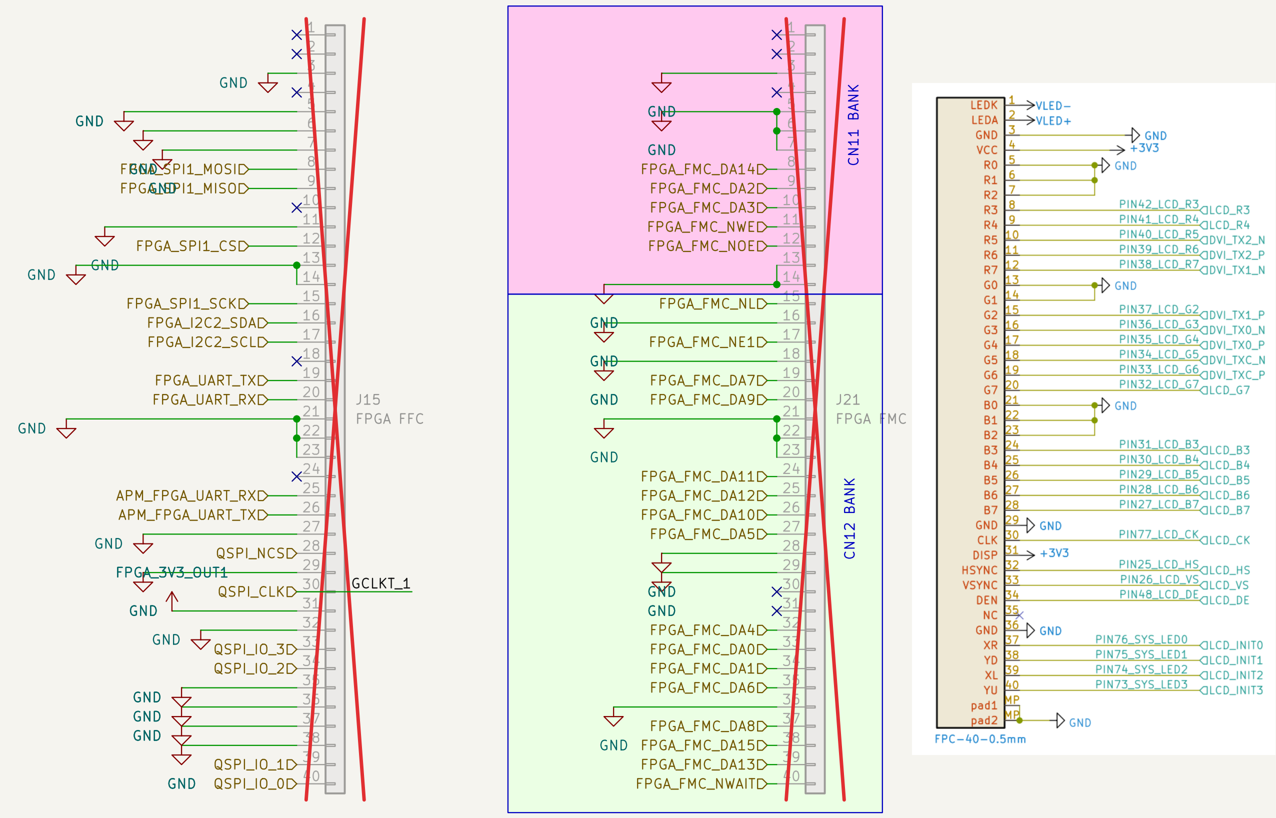
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 7/8

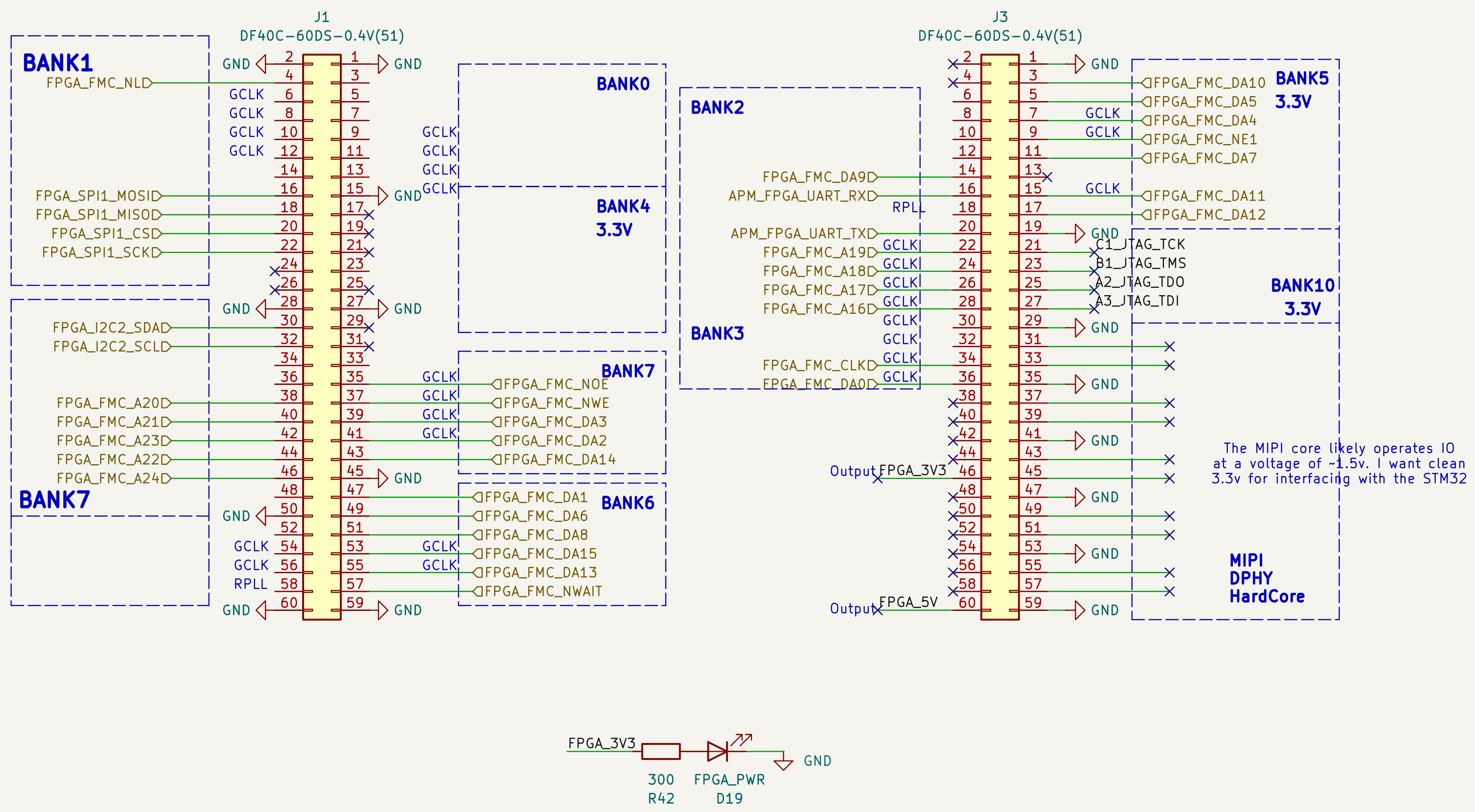
ACM FPGA Interface

Dual FPC STM32<->Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with:
    - 16 bit muxed data
    - and async FMC

configuration resulting in a 64MB shared address space between the STM32 & FPGA





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Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 8/8